

Supplementary material for “Scribble-supervised adaptation of SAM3 for text-prompted medical image segmentation”

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Supplementary per-structure results

Tables S1–S3 report per-structure results for each dataset. DSC, IoU, and NSD are shown as percentages; HD95 is reported in millimeters.

Table S1: Per-structure quantitative results on ACDC. Values are reported as mean $\pm$ standard deviation.

Structure	Method	DSC (%) $\uparrow$	HD95 $\downarrow$	IoU (%) $\uparrow$	NSD (%) $\uparrow$
left ventricle	BiomedParse-v2	96.04 $\pm$ 1.82	1.58 $\pm$ 0.16	92.44 $\pm$ 3.33	99.12 $\pm$ 0.86
	VoxTell	91.78 $\pm$ 4.97	1.28 $\pm$ 0.69	85.18 $\pm$ 8.15	99.23 $\pm$ 1.38
	SAT-Pro	93.88 $\pm$ 2.44	2.38 $\pm$ 0.48	88.55 $\pm$ 4.26	94.44 $\pm$ 2.69
	ScribFormer	90.78 $\pm$ 5.14	4.96 $\pm$ 5.56	83.49 $\pm$ 8.03	89.74 $\pm$ 7.17
	ScribbleBench	85.38 $\pm$ 15.32	86.03 $\pm$ 76.86	77.02 $\pm$ 19.14	82.15 $\pm$ 18.69
	EFFDNet	93.03 $\pm$ 5.58	7.76 $\pm$ 18.87	87.42 $\pm$ 8.71	94.51 $\pm$ 6.32
	SAM3-Scribble	93.95 $\pm$ 3.92	3.75 $\pm$ 6.66	88.82 $\pm$ 6.55	96.35 $\pm$ 4.42
	SAM3-Full	95.17 $\pm$ 3.94	2.17 $\pm$ 2.32	91.04 $\pm$ 6.67	97.57 $\pm$ 3.81
myocardium	BiomedParse-v2	92.21 $\pm$ 2.89	1.60 $\pm$ 0.24	85.67 $\pm$ 4.86	98.96 $\pm$ 1.09
	VoxTell	84.48 $\pm$ 4.08	1.20 $\pm$ 0.30	73.34 $\pm$ 6.05	99.30 $\pm$ 0.68
	SAT-Pro	87.87 $\pm$ 3.21	2.68 $\pm$ 0.77	78.50 $\pm$ 5.03	92.54 $\pm$ 3.21
	ScribFormer	81.86 $\pm$ 6.78	58.93 $\pm$ 81.93	69.79 $\pm$ 9.11	87.11 $\pm$ 8.18
	ScribbleBench	85.51 $\pm$ 4.61	70.13 $\pm$ 63.61	74.97 $\pm$ 6.96	88.46 $\pm$ 7.62
	EFFDNet	88.52 $\pm$ 2.24	4.24 $\pm$ 6.69	79.48 $\pm$ 3.61	94.45 $\pm$ 3.65
	SAM3-Scribble	90.36 $\pm$ 2.57	2.39 $\pm$ 2.10	82.52 $\pm$ 4.23	96.86 $\pm$ 2.63
	SAM3-Full	91.82 $\pm$ 1.92	2.27 $\pm$ 2.11	84.93 $\pm$ 3.30	98.10 $\pm$ 2.00
right ventricle	BiomedParse-v2	69.66 $\pm$ 30.08	14.61 $\pm$ 17.18	60.21 $\pm$ 30.87	72.06 $\pm$ 30.47
	VoxTell	84.83 $\pm$ 6.41	1.71 $\pm$ 0.69	74.18 $\pm$ 9.48	97.72 $\pm$ 1.52
	SAT-Pro	90.67 $\pm$ 3.66	3.23 $\pm$ 0.72	83.12 $\pm$ 5.79	88.10 $\pm$ 4.34
	ScribFormer	83.98 $\pm$ 10.93	7.33 $\pm$ 4.21	73.62 $\pm$ 13.76	81.26 $\pm$ 11.89
	ScribbleBench	83.27 $\pm$ 16.77	52.18 $\pm$ 68.32	74.09 $\pm$ 19.44	83.08 $\pm$ 16.89
	EFFDNet	87.15 $\pm$ 10.08	20.25 $\pm$ 37.81	78.40 $\pm$ 13.24	87.92 $\pm$ 10.80
	SAM3-Scribble	90.53 $\pm$ 7.78	4.27 $\pm$ 4.22	83.47 $\pm$ 10.92	93.06 $\pm$ 7.31
	SAM3-Full	91.79 $\pm$ 5.61	4.78 $\pm$ 3.98	85.30 $\pm$ 9.02	94.05 $\pm$ 5.61

Table S2: Per-structure quantitative results on BTCV_cervix. Values are reported as mean $\pm$ standard deviation.

Structure	Method	DSC (%) $\uparrow$	HD95 $\downarrow$	IoU (%) $\uparrow$	NSD (%) $\uparrow$
bladder	BiomedParse-v2	94.55 $\pm$ 1.43	3.05 $\pm$ 0.58	89.69 $\pm$ 2.56	85.96 $\pm$ 5.22
	VoxTell	94.89 $\pm$ 1.47	2.84 $\pm$ 0.61	90.31 $\pm$ 2.64	88.78 $\pm$ 5.74
	SAT-Pro	93.55 $\pm$ 2.04	3.69 $\pm$ 1.19	87.95 $\pm$ 3.55	83.82 $\pm$ 8.12
	ScribFormer	89.78 $\pm$ 5.80	16.14 $\pm$ 29.43	81.84 $\pm$ 8.93	65.99 $\pm$ 16.37
	ScribbleBench	94.49 $\pm$ 1.23	3.11 $\pm$ 0.80	89.58 $\pm$ 2.22	85.69 $\pm$ 5.59
	EFFDNet	89.27 $\pm$ 10.46	39.14 $\pm$ 65.46	82.01 $\pm$ 14.78	75.17 $\pm$ 21.20
	SAM3-Scribble	95.48 $\pm$ 0.90	2.45 $\pm$ 0.30	91.36 $\pm$ 1.65	92.02 $\pm$ 3.32
	SAM3-Full	95.49 $\pm$ 0.95	2.49 $\pm$ 0.33	91.39 $\pm$ 1.74	92.22 $\pm$ 3.20
rectum	BiomedParse-v2	0.00 $\pm$ 0.00	277.53 $\pm$ 85.55	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
	VoxTell	81.79 $\pm$ 6.67	21.43 $\pm$ 23.22	69.72 $\pm$ 9.29	72.97 $\pm$ 8.75
	SAT-Pro	69.70 $\pm$ 12.23	43.02 $\pm$ 61.17	54.59 $\pm$ 14.14	58.32 $\pm$ 10.40
	ScribFormer	59.31 $\pm$ 13.41	47.04 $\pm$ 41.73	43.39 $\pm$ 15.52	38.50 $\pm$ 12.88
	ScribbleBench	71.02 $\pm$ 11.90	30.15 $\pm$ 24.87	56.36 $\pm$ 14.09	62.90 $\pm$ 12.77
	EFFDNet	65.52 $\pm$ 9.02	39.02 $\pm$ 39.97	49.43 $\pm$ 10.54	48.52 $\pm$ 7.99
	SAM3-Scribble	73.54 $\pm$ 9.32	33.50 $\pm$ 19.60	59.07 $\pm$ 12.46	65.75 $\pm$ 8.41
	SAM3-Full	73.15 $\pm$ 10.93	33.18 $\pm$ 23.37	58.85 $\pm$ 13.86	65.76 $\pm$ 9.82
small bowel	BiomedParse-v2	36.24 $\pm$ 15.11	89.71 $\pm$ 36.25	23.04 $\pm$ 11.77	21.61 $\pm$ 9.48
	VoxTell	67.64 $\pm$ 6.50	59.01 $\pm$ 38.07	51.48 $\pm$ 7.49	44.66 $\pm$ 5.48
	SAT-Pro	52.51 $\pm$ 10.64	62.76 $\pm$ 22.48	36.22 $\pm$ 10.21	33.56 $\pm$ 10.58
	ScribFormer	41.29 $\pm$ 15.69	50.64 $\pm$ 28.36	27.09 $\pm$ 13.11	25.05 $\pm$ 7.36
	ScribbleBench	54.96 $\pm$ 9.19	44.15 $\pm$ 32.07	38.43 $\pm$ 8.56	37.24 $\pm$ 8.23
	EFFDNet	52.67 $\pm$ 13.32	45.85 $\pm$ 22.97	36.98 $\pm$ 13.73	33.11 $\pm$ 12.47
	SAM3-Scribble	63.05 $\pm$ 10.02	35.31 $\pm$ 19.01	46.87 $\pm$ 11.33	47.50 $\pm$ 5.94
	SAM3-Full	53.15 $\pm$ 9.47	39.73 $\pm$ 15.95	36.81 $\pm$ 9.61	39.88 $\pm$ 6.10
uterus	BiomedParse-v2	14.92 $\pm$ 16.12	53.39 $\pm$ 15.58	8.79 $\pm$ 10.06	9.98 $\pm$ 11.46
	VoxTell	84.48 $\pm$ 4.93	7.03 $\pm$ 4.32	73.43 $\pm$ 7.11	71.28 $\pm$ 13.13
	SAT-Pro	67.69 $\pm$ 7.76	24.10 $\pm$ 6.87	51.61 $\pm$ 9.16	44.25 $\pm$ 10.58
	ScribFormer	66.17 $\pm$ 5.07	23.11 $\pm$ 6.73	49.62 $\pm$ 5.63	40.11 $\pm$ 8.14
	ScribbleBench	78.27 $\pm$ 4.40	19.27 $\pm$ 11.54	64.51 $\pm$ 5.98	62.25 $\pm$ 9.87
	EFFDNet	67.57 $\pm$ 14.21	35.50 $\pm$ 39.11	52.50 $\pm$ 13.81	49.56 $\pm$ 14.56
	SAM3-Scribble	75.30 $\pm$ 6.15	16.79 $\pm$ 7.10	60.75 $\pm$ 7.60	58.91 $\pm$ 8.19
	SAM3-Full	78.40 $\pm$ 7.50	13.94 $\pm$ 9.21	65.09 $\pm$ 10.12	67.07 $\pm$ 12.83

Table S3: Per-structure quantitative results on PROMISE12. Values are reported as mean $\pm$ standard deviation.

Structure	Method	DSC (%) $\uparrow$	HD95 $\downarrow$	IoU (%) $\uparrow$	NSD (%) $\uparrow$
prostate	BiomedParse-v2	85.97 $\pm$ 6.31	14.78 $\pm$ 13.63	75.90 $\pm$ 9.29	76.01 $\pm$ 11.93
	VoxTell	80.81 $\pm$ 13.97	27.61 $\pm$ 31.18	69.87 $\pm$ 17.73	71.06 $\pm$ 18.87
	SAT-Pro	88.02 $\pm$ 3.73	4.15 $\pm$ 1.21	78.79 $\pm$ 5.80	79.82 $\pm$ 8.53
	ScribFormer	78.13 $\pm$ 8.06	12.91 $\pm$ 9.14	64.78 $\pm$ 10.51	59.78 $\pm$ 13.46
	ScribbleBench	88.90 $\pm$ 2.90	4.79 $\pm$ 1.78	80.14 $\pm$ 4.62	82.01 $\pm$ 5.92
	EFFDNet	87.69 $\pm$ 3.71	10.67 $\pm$ 17.39	78.27 $\pm$ 5.74	79.19 $\pm$ 8.20
	SAM3-Scribble	90.36 $\pm$ 3.45	5.95 $\pm$ 6.22	82.58 $\pm$ 5.56	84.99 $\pm$ 7.29
	SAM3-Full	91.53 $\pm$ 2.98	6.05 $\pm$ 6.76	84.52 $\pm$ 4.95	87.39 $\pm$ 7.38

Supplementary ablation and sensitivity results

Tables S4–S8 provide the complete macro-averaged ablation and threshold sensitivity results. DSC, HD95, IoU, and NSD are reported in this order; DSC, IoU, and NSD are percentages, whereas HD95 is reported in millimeters.

Table S4: Box and GIoU regression loss ablation. All variants use the no-geometric matcher, partial mask loss, hybrid LoRA, and the same validation-based checkpoint selection.

Variant	box/GIoU loss	DSC (%) ↑	HD95 ↓	IoU (%) ↑	NSD (%) ↑
No geometric regression	0/0	86.27	10.48	77.35	82.15
Full geometric regression	5/2	85.86	10.27	76.75	81.65
Reduced geometric regression	1/1	85.95	11.32	76.99	81.87

Table S5: Hungarian matcher cost ablation. All variants use partial mask loss, disabled box/GIoU regression losses, hybrid LoRA, and the same validation-based checkpoint selection.

Variant	cls/box/GIoU cost	DSC (%) ↑	HD95 ↓	IoU (%) ↑	NSD (%) ↑
Original matcher	2/5/2	86.19	10.55	77.24	81.74
No geometric matcher	2/0/0	86.27	10.48	77.35	82.15
Reduced geometric matcher	2/1/1	86.12	10.65	77.18	81.83
Rebalanced matcher	5/1/1	85.83	10.90	76.80	81.69

Table S6: LoRA rank ablation. All variants use the proposed scribble-compatible objective and hybrid LoRA scope.

Variant	rank/alpha	DSC (%) ↑	HD95 ↓	IoU (%) ↑	NSD (%) ↑
L1 Low rank	4/8	86.33	10.45	77.35	81.70
L0 Proposed	8/16	86.27	10.48	77.35	82.15
L2 High rank	16/32	85.66	11.38	76.58	81.19

Table S7: LoRA scope ablation. Rank is fixed to 8 and alpha to 16 when LoRA modules are present. The segmentation and scoring heads are fine-tuned in all variants.

Variant	LoRA scope	DSC (%) $\uparrow$	HD95 $\downarrow$	IoU (%) $\uparrow$	NSD (%) $\uparrow$
Hybrid LoRA	vision/text/geometry/DETR	86.27	10.48	77.35	82.15
Vision-only LoRA	vision encoder	85.83	10.47	76.78	81.01
DETR-only LoRA	DETR encoder/decoder	82.82	12.79	73.36	78.08
No LoRA	none	62.91	39.60	51.24	53.88

Table S8: Inference confidence threshold sensitivity. Each row is macro-averaged over ACDC, BTCV_cervix, and PROMISE12 using the same trained checkpoint for the corresponding model.

Model	Threshold	DSC (%) $\uparrow$	HD95 $\downarrow$	IoU (%) $\uparrow$	NSD (%) $\uparrow$
SAM3-Scribble	0.3	85.79	11.74	76.61	80.95
SAM3-Scribble	0.5	86.14	10.99	77.12	81.61
SAM3-Scribble	0.7	86.27	10.48	77.35	82.15
SAM3-Scribble	0.9	85.91	9.99	77.03	81.89
SAM3-Full	0.3	86.32	12.63	77.77	83.06
SAM3-Full	0.5	86.53	11.28	78.12	83.36
SAM3-Full	0.7	86.50	10.49	78.22	83.40
SAM3-Full	0.9	66.27	20.94	57.03	60.80

Table S9: Paired Wilcoxon signed-rank tests on per-case DSC scores. Positive Δ indicates that SAM3-Scribble is better than the compared method. p_{Holm} is corrected within the DSC family across all listed comparisons.

Dataset	Comparison	n	Δ DSC (pp)	p	p_{Holm}
ACDC	SAM3-Scribble vs ScribbleBench	30	6.89	< 0.001	$< 0.001^*$
ACDC	SAM3-Scribble vs EFFDNet	30	1.17	< 0.001	0.002^*
ACDC	SAM3-Scribble vs ScribFormer	30	6.07	< 0.001	$< 0.001^*$
ACDC	SAM3-Scribble vs SAM3-Full	30	-1.32	< 0.001	0.002^*
ACDC	SAM3-Scribble vs SAT-Pro	30	0.81	0.008	0.046^*
BTCV_cervix	SAM3-Scribble vs ScribbleBench	6	2.16	0.156	0.312
BTCV_cervix	SAM3-Scribble vs EFFDNet	6	10.68	0.031	0.156
BTCV_cervix	SAM3-Scribble vs ScribFormer	6	12.70	0.031	0.156
BTCV_cervix	SAM3-Scribble vs SAM3-Full	6	1.80	0.562	0.562
BTCV_cervix	SAM3-Scribble vs SAT-Pro	6	5.98	0.031	0.156
PROMISE12	SAM3-Scribble vs ScribbleBench	30	1.45	< 0.001	0.005^*
PROMISE12	SAM3-Scribble vs EFFDNet	30	2.67	< 0.001	$< 0.001^*$
PROMISE12	SAM3-Scribble vs ScribFormer	30	12.23	< 0.001	$< 0.001^*$
PROMISE12	SAM3-Scribble vs SAM3-Full	30	-1.17	0.003	0.019^*
PROMISE12	SAM3-Scribble vs SAT-Pro	30	2.34	0.002	0.016^*

Asterisks denote comparisons that remain significant at $\alpha = 0.05$ after Holm correction within the DSC family.